Last mile data analysis and visualization

Course Project - 2

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by

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List of Tables

3.1 Categorization of income 8
3.2 Categorization of age 8
3.3 Mode categorization for all the cities 10
3.4 Final last mile mode categorization 11
3.5 Computation of values of the independent variables 12
3.6 Descriptive Statistics for Bengaluru Last-Mile Variables 13
3.7 Descriptive Statistics for Delhi Last-Mile Variables 13
3.8 Descriptive Statistics for Nagpur Last-Mile Variables 13
3.9 City-wise Chi-square results: Mode choice vs. demographics 14
3.10 City-wise one-way ANOVA (Last-mile travel time by mode) 15
3.11 Mean last-mile travel time (minutes) by mode and city 16
3.12 City-wise Pearson correlation between last-mile time (TT_LM) and cost (TC_LM) 17

4.1 Nagpur – Station-Level Last-Mile Performance Summary 25
4.2 Delhi – Station-Level Last-Mile Performance Summary 26
4.3 Bengaluru – Station-Level Last-Mile Performance Summary 27
4.4 Comparative Summary of Accessibility Indicators 27

Chapter 1

Introduction

Urban mobility systems are increasingly defined by their efficiency in connecting people seamlessly between different modes of transport. Among these connections, the last-mile link—the segment bridging the nearest public transport node and the commuter's final destination—plays a decisive role in determining the overall accessibility, reliability, and attractiveness of the public transport system (iTrans Consulting 2022; Litman 2019). Despite being the shortest physical segment of a trip, it often represents the largest perceptual and logistical barrier to achieving fully integrated multi-modal travel (Soman & Gadepalli 2021). In rapidly urbanizing Indian cities such as Bengaluru, Delhi, and Nagpur, addressing the last-mile problem has become essential for improving transit ridership, reducing congestion, and advancing sustainable urban transport goals (Ministry of Housing and Urban Affairs 2015; Ramachandra & Bharath 2020).

The primary objective of this project is to analyze and visualize the behavioural and spatial patterns associated with last-mile connectivity across the three cities, using real-world metro survey data. The study aims to uncover how commuters' demographic characteristics (such as income, age, and gender) and trip attributes (like travel time, distance, and cost) influence their mode choice for the last-mile connection (Pawar & Yadav 2020; Raja & Gadepalli 2022). The project also explores the extent to which physical accessibility, urban form, and infrastructure provision shape these mode choices, thereby identifying stations or zones that face higher dependence on motorized or intermediate feeder modes (Gadepalli 2016).

This project adopts a data-driven analytical approach integrating two major tools—Power BI and QGIS—to transform raw survey data into meaningful insights. Power BI enables dynamic visualization and interpretation of behavioural variables such as travel time, waiting time, cost, and demographic dependencies through interactive dashboards (Microsoft Corporation 2023). On the other hand, QGIS facilitates spatial mapping and analysis of accessibility indicators—such as dominant last-mile mode, percentage of walk trips, and average last-mile travel time—at the metro-station level (QGIS Development Team 2022). Through geospatial representation, QGIS identifies areas that are well connected by non-motorized transport (NMT) as well as those with poor pedestrian access or excessive reliance on intermediate public transport (IPT).

By combining these two tools, the project provides a holistic understanding of both behavioural and spatial dimensions of last-mile travel. The analysis focuses on three key outcomes:

- Identifying the most common last-mile modes and their variation across socio-economic groups.
- Measuring accessibility performance through travel-time and walk-share indicators.
- Comparing inter-city differences in connectivity and identifying high non-walk dependency zones.

The importance of this study lies in its potential to inform policy and design interventions aimed at improving metro accessibility (Gadepalli & Mahendra 2018). With the rapid expansion of metro systems in India, planners face the challenge of ensuring that station access remains equitable, convenient, and sustainable. Findings from this project can guide infrastructure prioritization, such as developing shaded and continuous pedestrian paths, integrating IPT services through formalized feeder routes, and promoting bicycle-based access solutions (Institute for Transportation and Development Policy (ITDP India) 2020). Furthermore, insights drawn from demographic and income-based segmentation highlight the need for inclusive planning that considers affordability, gender safety, and universal design (World Bank 2022).

In addition to the analytical outputs, this project demonstrates the relevance of data visualization as a planning support tool. The Power BI dashboard serves as a communication interface for researchers, policymakers, and the public, enabling transparent decision-making through visual analytics (Zhou & Wang 2021). The QGIS maps, on the other hand, translate data into spatial narratives, making it easier to pinpoint geographic gaps in accessibility. Together, these approaches reinforce the idea that effective last-mile solutions must be both data-informed and user-centric ((ITRA) 2023).

Overall, the project contributes to a broader understanding of how behavioural preferences, socio-economic factors, and spatial infrastructure jointly determine last-mile connectivity in Indian metro systems. It also underscores the potential of integrating digital analytics and geospatial technologies in transport planning education and practice. The subsequent chapters describe the literature background, methodology, analytical procedures in Power BI and QGIS, and evidence-based recommendations derived from the findings.

Chapter 2

Literature review

2.1 Literature Review on Innovations in Last-Mile Connectivity

2.1.1 Concept and Evolution of LMC

The contemporary concept of Last-Mile Connectivity (LMC) is increasingly viewed as the core enabler of inclusive and sustainable public transit. An effective LMC system determines whether the metro network is merely an infrastructure project or a societal asset accessible to all. By solving the *last-mile problem*, LMC mitigates reliance on private vehicles for short trips, thereby reducing traffic congestion and carbon emissions. Functionally, it expands the transit network's effective coverage, ensuring that marginalized populations and residents of low-density areas can access high-capacity transit. Thus, LMC is a critical lever for achieving both equity and sustainability goals in urban mobility.

2.1.2 Shared Mobility and Integration Approaches

The challenge of bridging the transit gap—the distance between a user's origin/destination and the metro station—is increasingly addressed through the integration of public transport with shared mobility modes. (Hoen *et al.* 2024) provide a critical analysis of how modes like bike-sharing, e-scooters, and pooled rides enhance LMC. Their work underscores that success hinges on seamless digital and physical integration. Digital integration involves unifying payment and route planning via a single platform (Mobility-as-a-Service, or MaaS), while physical integration requires co-located infrastructure for parking, pick-up/drop-off, and easy transfer at metro stations. The authors stress that these shared modes offer flexibility and coverage that fixed-route feeders cannot, effectively expanding the metro's catchment area and increasing overall system ridership by improving the first- and last-mile experience.

2.1.3 Operational Innovations: MARTA Reach Pilot

(Van Hentenryck *et al.* 2023) offer a practical case study in LMC innovation through the MARTA Reach pilot in Atlanta, which deployed an on-demand multimodal transit system. This approach replaces fixed-route feeder services in low-density areas with dynamically routed shared vans and shuttles, driven by real-time user requests

via a mobile app. The study highlights the successful application of advanced routing and dispatch algorithms to optimize vehicle utilization while maintaining acceptable wait and travel times for passengers. A core finding is the concept of "flexible service boundaries," where the on-demand zone is strategically defined to complement, not compete with, the existing fixed-route metro and bus network. The pilot demonstrates that on-demand systems can provide a cost-effective, high-coverage access solution in challenging service areas.

2.1.4 Analytical Tools and Methods

A robust analysis of LMC systems relies on a suite of sophisticated tools. Simulation models (e.g., agent-based or discrete-event) are crucial for forecasting the impact of new mobility services on network performance. Accessibility mapping—often leveraging Geographic Information Systems (GIS) and isochrone analysis—quantifies the service area and equity of access before and after LMC interventions. Ridership tracking via smart card or app data provides essential behavioral insights and validates service demand. Finally, cost–benefit frameworks are necessary to justify investments, moving beyond simple operational costs to include social benefits like reduced emissions and increased mobility.

2.1.5 Key Findings, Research Gaps, and Relevance

The literature confirms a powerful shift toward user-centric, integrated, and flexible LMC solutions—moving from static infrastructure toward dynamic, digitally enabled mobility networks. The insights from (Hoen *et al.* 2024; Van Hentenryck *et al.* 2023) provide direct, actionable evidence for how shared and on-demand services can transform metro access.

Despite these advances, significant research gaps remain: limited real-world validation in dense, complex developing contexts like India; a lack of standardized equity indicators to assess fairness in service delivery; and weak data integration across public and private modes. For the Bengaluru, Delhi, and Nagpur project, the key challenge is transferability. The operational successes of pilots like MARTA Reach must be adapted and validated against the unique behavioral patterns, varied regulatory landscapes, and high passenger volumes of Indian metro systems, demanding a focus on diagnostic work grounded in local data.

Chapter 3

Methodology

3.1 Data Cleaning

3.1.1 Last-mile travel time

A few respondents across the three city datasets did not provide valid entries for last-mile travel time. Since this variable is fundamental to most subsequent analyses, such records were excluded to maintain analytical consistency. The number of removed observations is summarized below:

- **Bangalore** — 2 respondents
- **Delhi** — 1 respondent
- **Nagpur** — 71 respondents

These omissions represent a very small proportion of the total dataset and therefore do not affect the overall validity or representativeness of the results.

3.1.2 Same origin–destination (OD) pair

Some respondents reported the same location for both their trip origin and destination, implying no actual travel took place. Such cases were excluded to ensure the analysis reflected genuine last-mile travel behavior. The number of excluded records for each city is shown below:

- **Bangalore** — 0 respondents
- **Delhi** — 67 respondents
- **Nagpur** — 2 respondents

Excluding these entries helps maintain the logical integrity of the dataset and avoids distortion of travel time and distance distributions.

3.1.3 Last-mile travel distance

A small subset of respondents across the cities had missing, zero, or implausibly high values for last-mile travel distance. These were carefully reviewed and handled to preserve both data quality and representativeness. The summary is as follows:

- **Bangalore** — Although approximately 450 entries showed missing or inconsistent distance values, they were retained to maintain representativeness of the dataset ($n = 2205$). Removing them would have disproportionately reduced the sample size.
- **Delhi** — 125 entries with invalid or missing distance values (about 3.4% of $n = 3708$) were removed. Additionally, 8 outliers with unrealistically high last-mile distances (exceeding 100 km) were excluded to ensure realistic distribution of travel distances.
- **Nagpur** — 86 invalid or missing entries (around 3.3% of $n = 2573$) were excluded after confirming that their removal had no effect on overall representativeness.

This selective cleaning approach ensured that the final dataset was free from inconsistencies and outliers while maintaining its overall balance and statistical reliability.

3.2 Data processing

3.2.1 Income categorization

According to (Ministry of Housing and Urban Affairs, Government of India 2024) and (Ministry of Housing and Urban Affairs 2024), the official income-based categorization defines households with a monthly income of $\leq$ 50,000 as Low Income (LI), those with 50,000 $<$ income $\leq$ 75,000 as Middle Income (MI), and those above 75,000 as High Income (HI).

However, since the income distribution in our dataset differs from these standard ranges, the classification thresholds were slightly adjusted to better align with the data's actual spread and to ensure meaningful segmentation for analysis. The final categorization adopted for this study is presented in Table 3.2. Records where respondents chose not to disclose their income were retained under a separate Refused/NA category to avoid data loss and maintain completeness.

Table 3.1: Categorization of income

Income category	Official range (INR)	Selected range for our analysis (INR)
Low income (LI)	Monthly income $\leq$ 50k	Monthly income $\leq$ 60k
Medium income (MI)	50k $<$ Monthly income $\leq$ 75k	60k $<$ Monthly income $\leq$ 80k
High income (HI)	Monthly income $>$ 75k	Monthly income $>$ 75k

3.2.2 Age group categorization

According to the (Office of the Registrar General & Census Commissioner, India 2011) report we can classify respondents into the following categories - Child $\rightarrow$ 0 - 14 years ; Adult $\rightarrow$ 15 - 59 years and Old $\rightarrow$ 60 and above. However, the survey age entries in our dataset are not recorded uniformly (mix of single ages and uneven brackets), we adopted a set of analysis-friendly age bins. The final categorisation used in this study is shown in Table 3.2

Table 3.2: Categorization of age

Age group	Official range (years)	Selected range for our analysis (years)
Child	Age $\leq$ 14	Age $\leq$ 18
Adult	14 $<$ Age $\leq$ 59	19 $<$ Monthly income $\leq$ 59
Old	Age $>$ 60	Age $>$ 60

3.2.3 OD pair formation

In the given dataset, we have survey station name, survey location i.e. whether the survey was conducted at the entry gate or exit gate and origin or destination of the respondents in a single column. From these available information we have mapped the OD pair in the following way :

If survey was conducted at the entry gate :

- Origin : Survey station
- Destination : Respondents response from the OD column

If survey was conducted at the exit gate :

- Origin : Respondents response from the OD column
- Destination : Survey station

3.2.4 Last mile mode categorization

For all the cities respondents have used more than 10 modes as egress mode. But, for our analysis it is not desirable and advisable to use them individually to observe the last mile mode share pattern. Thus we convert them into the following categories as per standard literature like (Djavadian *et al.* 2021), (Z. Liu & J. Liu 2023), (Widiastuti 2024), (Thorhauge *et al.* 2022) etc..

Table 3.3: Mode categorization for all the cities

Categorized mode	Bengaluru	Delhi	Nagpur
NMT	• Walk • Cycle	• Walk • Cycle	• Walk • Cycle
Private vehicle (PV)	• Four wheeler (self drive) • Two wheeler (self drive)	• Four wheeler (self drive) • Two wheeler (self drive)	• Four wheeler (self drive) • Two wheeler (self drive)
Intermediate para transit (IPT)	• Auto (app based) • Auto (hail regular) • Auto (shared) • Public bicycle sharing (app based) • Car pooling • Taxi (app based) • Two wheeler (app based)	• Company transport • Auto (hail regular) • Auto (shared) • Cycle rickshaw • Car pooling • Shared taxi • DTC bus • E auto • E rickshaw • Gramin sewa • Metro feeder bus • RTV	• Bus • Auto (hail regular) • Auto (shared) • Car pooling • Shared taxi

3.2.5 Final last mile mode categorization

As we have discussed previously in the section 3.2.4, we have categorized all the available last mile mode into three categories i.e. NMT, IPT and private vehicles (PV). Since, for our further analysis we are interested in last mile mode choice of the respondents from the metro station only. And in our given dataset we have two legs for primary mode as well. Now, if for primary leg 2, if they are not using metro then we have combined them with last mile modes. On the other hand if for primary leg 1, if they are not using metro then we have combined them with first mile mode. The detailed last mile mode combination is shown in the table below -

Table 3.4: Final last mile mode categorization

Primary leg 2	Primary leg 2 category	Last mile categories	Final last mile mode combination
Walking	NMT	NMT	NMT
		IPT	IPT + NMT
		PV	PV
Two wheeler or four wheeler (self drive)	Private vehicle (PV)	NMT	PV
Bus	Bus	NMT	Bus + NMT
		IPT	Bus + IPT
		PV	Bus + PV
Any other mode apart from NMT, PV and Bus like auto, e-hailing services	IPT	NMT	IPT + NMT
		IPT	IPT
		PV	IPT + PV

→ So the final last mile mode combinations are as follows

- Bus + NMT → + or
- Bus + IPT → + or
- Bus + PV → + or
- IPT + PV → or + or
- IPT + NMT → or + or
- NMT only → or
- IPT only → or
- PV only → or

3.2.6 Numerical computation

For the visualization of our transformed data and draw some inference from them, we need to have the values some important variables like travel time, travel cost, waiting time etc. for the last mile mode. And the values are not given directly. So, we have computed them as shown in the table below -

Table 3.5: Computation of values of the independent variables

Sl. no	Independent variables	Computation of numeric values
1	Last mile travel time (TT_{LM}) (minutes)	Travel time of last mile leg 1 + last mile leg 2
2	Last mile travel cost (TC_{LM}) (INR)	Fixed travel cost of last mile leg 1 + last mile leg 2 + (monthly travel cost of last mile leg 1 or leg 2) /30
3	Last mile travel distance (TD_{LM}) (Km)	Last mile : Leg 1 travel distance in km + leg 2 travel distance in km + (leg 1 travel distance in meter + leg 2 travel distance in meter) /1000
4	Last mile waiting time (WT_{LM}) (minutes)	Last mile leg 1 waiting time + last mile leg 2 waiting time
5	Vehicle ownership (nos)	Nos of 2 wheeler own by the respondent + nos of four wheeler own by the respondent

We did not include the primary mode leg 2 details in the above computation as for all the cities and across all the respondents the data is not available. So to maintain the uniformity of our analysis we have included only values from the last mile modes.

3.3 Data analysis

To understand the characteristics of last-mile travel, the analysis was divided into two parts: descriptive analysis and statistical testing.

3.3.1 Descriptive Analysis

This section summarizes the overall distribution of last-mile variables such as travel time, cost, and distance. Key statistical measures including mean, median, and mode were used to describe central tendency, while standard deviation and variance captured the spread of data. Frequency histograms were also generated to visualize mode shares, travel times, and costs across the three cities.

Table 3.6: Descriptive Statistics for Bengaluru Last-Mile Variables

Variable	Count	Mean	Median	Mode	Std	Var	Min	Q1	Q3	Max
TT_LM	2203	10.7	10	10	6.3	40.0	0	6.0	15.0	100
TC_LM	2203	10.6	0	0	37.9	1435.1	0	0.0	0.0	500
TD_LM	2203	1.1	0.75	0.75	1.9	3.5	0	0.2	1.0	35

Table 3.7: Descriptive Statistics for Delhi Last-Mile Variables

Variable	Count	Mean	Median	Mode	Std	Var	Min	Q1	Q3	Max
TT_LM	3507	15.1	10	10	13.0	169.6	0	7.0	20.0	250
TC_LM	3507	5.6	0	0	13.1	172.5	0	0.0	10.0	310
TD_LM	3507	2.7	0.6	0.5	17.5	305.8	0	0.3	2.1	501

Table 3.8: Descriptive Statistics for Nagpur Last-Mile Variables

Variable	Count	Mean	Median	Mode	Std	Var	Min	Q1	Q3	Max
TT_LM	2500	13.6	10	10	10.4	108.4	1	10.0	15.0	130
TC_LM	2500	4.2	0	0	10.3	106.5	0	0.0	4.0	120
TD_LM	2500	2.0	0.9	0.5	4.1	16.7	0.2	0.5	2.0	75.5

3.3.2 Statistical Testing

To explore relationships and differences between variables, statistical tests were conducted. The chi-square test examined associations between mode choice and demographic attributes, while one-way ANOVA assessed whether average travel times significantly differed across modes. Correlation analysis was also performed to evaluate the relationship between travel cost and travel time.

Chi-square test: Mode choice vs demographic attributes

The Chi-Square (χ^2) Test of Independence was applied to assess whether last-mile mode choice (collapsed into NMT, IPT, PV, BUS, and MIXED) is associated with demographic variables — gender, age, and income group.

This non-parametric test determines if observed mode-choice distributions differ significantly from those expected under independence.

The null hypothesis (H_0) assumes that mode choice is independent of demographic factors, while the alternative hypothesis (H_1) indicates dependence.

Association strength was measured using Cramer's V, where values below 0.1 represent a negligible relationship.

Note: Before applying the Chi-Square (χ^2) Test, low-frequency multimodal combinations were merged into a single MIXED category to ensure statistical reliability (expected frequency ≥ 5). The MIXED group includes combined trips such as BUS + IPT, BUS + NMT, BUS + PV, IPT + NMT, and PV + IPT, representing integrated public, intermediate, and non-motorized travel. This produced five consistent analytical classes — NMT, IPT, PV, BUS, and MIXED — enabling uniform comparison across all cities.

City-Wise Results : We have tabulated the city wise results as shown in the table below

Table 3.9: City-wise Chi-square results: Mode choice vs. demographics

City (N)	**Attribute**	$\boldsymbol{\chi^2}$	p**-value**	V	**Interpretation**
Nagpur (2500)	Gender	1.76	0.78	0.027	Independent
	Age	7.81	0.45	0.040	Independent
	Income	37.48	0.00018^{**}	0.071	Weak, significant
Bengaluru (2203)	Gender	1.27	0.74	0.024	Independent
	Age	4.69	0.86	0.027	Independent
	Income	44.62	0.000001^{**}	0.082	Weak, significant
Delhi (3507)	Gender	8.85	0.18	0.035	Independent
	Age	17.94	0.036^{*}	0.040	Weak, significant
	Income	131.55	$< 0.001^{**}$	0.109	Weak, significant

Notes: Where, N is the nos of respondents, V = Cramer's V (effect size). Thresholds (rule-of-thumb): $V < 0.10$ small/weak, $0.10-0.30$ moderate. Significance marks: $^{*}\, p < 0.05$, $^{**}\, p < 0.01$.

Cross-City Summary: Across all cities, gender is independent of mode choice ($p > 0.05$).
Age is independent in Nagpur and Bengaluru but weakly significant in Delhi.

Income group shows weak statistical significance in all three cities, with the strongest (yet still small) effect in Delhi.
In every case, Cramer's V < 0.1, confirming that these relationships—though occasionally significant—are practically negligible.

ANOVA: Differences in travel times across modes

A one-way Analysis of Variance (ANOVA) was conducted to test whether average last-mile travel times differed significantly across modes—NMT, IPT, PV, BUS, and MIXED. Separate analyses were performed for Nagpur, Bengaluru, and Delhi. The test evaluated mean differences at a significance level of $\alpha = 0.05$, and effect sizes were estimated using η^2 (eta squared) and ω^2 (omega squared), where values above 0.14 denote large effects, 0.06 medium, and 0.01 small.

City-Wise Results : We have tabulated the city wise results as shown in the tables below

Table 3.10: City-wise one-way ANOVA (Last-mile travel time by mode)

City (N)	F	**df (between, within)**	p**-value**	$\boldsymbol{\eta^2}$	$\boldsymbol{\omega^2}$	**Interpretation / Post-hoc (Tukey)**
Nagpur (2500)	243.19	(4, 2495)	< 0.001	0.281	0.279	Large effect; MIXED, BUS ↑ longer than others; NMT shortest
Bengaluru (2203)	65.80	(3, 2201)	< 0.001	0.082	0.081	Moderate effect; IPT, PV ↑ longer than NMT, MIXED
Delhi (3507)	502.03	(3, 3704)	< 0.001	0.289	0.288	Large effect; MIXED ↑ longest; NMT, PV shortest and similar

Notes: One-way ANOVA for last-mile travel time across modes (NMT, IPT, PV, BUS, MIXED where available). Effect-size guide (rule-of-thumb): $\eta^2, \omega^2 \approx 0.01$ small, ≈ 0.06 moderate, $\gtrsim 0.14$ large.

Table 3.11: Mean last-mile travel time (minutes) by mode and city

City	NMT	IPT	PV	BUS	MIXED
Nagpur	10.94	13.92	15.13	23.20	26.21
Bengaluru	9.89	14.97	17.20	—	10.79
Delhi	10.34	17.42	10.29	—	28.31

Notes: Means are in minutes. "—" indicates the mode was not analysed/reported for that city. Post-hoc (Tukey) highlights: Nagpur—MIXED, BUS > others ($p < 0.001$), NMT shortest; Bengaluru—IPT, PV > NMT, MIXED ($p < 0.001$); Delhi—MIXED longest ($p < 0.001$), NMT and PV shortest and statistically similar.

Cross-City Summary : The one-way ANOVA results across Nagpur, Bengaluru, and Delhi indicate statistically significant differences in mean travel times across last-mile modes ($p < 0.001$).

Mixed and bus-based modes consistently exhibit the highest travel times due to multimodal transfers and extended route complexity, while non-motorized modes (NMT) record the shortest and most consistent durations.

Effect sizes (η^2, ω^2) show large modal impacts in Delhi and Nagpur and a moderate effect in Bengaluru.

In Bengaluru, the standalone BUS group was excluded from ANOVA due to insufficient observations ($n < 5$); however, bus-linked trips were retained within the MIXED category, ensuring their representation in the analysis.

Overall, the findings confirm that mode type exerts a strong and statistically significant influence on last-mile travel time variability.

3.3.3 Correlation: Relationship between last-mile cost and time

To examine the degree of association between monetary expenditure and temporal effort in last-mile travel, a bivariate correlation analysis was performed between last-mile travel time (TT_LM) and last-mile cost (TC_LM) for each city. The Pearson product–moment correlation coefficient (r) was computed, which quantifies the linear relationship between two continuous variables. Values of r range between -1 and $+1$, where positive values denote direct proportionality (as one variable increases, the other tends to increase), and negative values indicate an inverse relationship. The significance of each correlation was tested using a two-tailed p-value with a 95% confidence level.

Interpretation: All three correlations are positive and statistically significant at $\alpha = 0.05$, indicating that respondents who spend more time on the last-mile segment also tend to incur higher out-of-pocket costs. This reinforces the expected behavioural consistency where longer access or egress distances translate to higher expenditure, especially for motorised or multi-modal trips.

- **Bengaluru ($r = 0.38$):** Displays a moderate relationship, implying a relatively flatter cost–time gradient, possibly due to higher walk and shared-auto shares that mitigate cost escalation despite longer travel times.
- **Delhi ($r = 0.52$):** Shows a moderate-to-strong relationship, suggesting that even with a wide availability of modes, users still experience proportionate increases in cost with longer durations, potentially due to congestion and fare-based pricing.
- **Nagpur ($r = 0.61$):** Exhibits the strongest positive relationship, reflecting a pronounced dependency between time and cost. This may indicate higher unit costs of travel (e.g., per-minute or per-kilometre rates in IPT and shared modes) or lower modal diversity, leading to limited low-cost alternatives.

Table 3.12: City-wise Pearson correlation between last-mile time (TT_LM) and cost (TC_LM)

City	N	**Pearson r (95% CI)**	p**-value**
Bengaluru	2203	0.377 (0.341, 0.412)	1.98×10^{-75}
Delhi	3507	0.523 (0.499, 0.547)	7.51×10^{-246}
Nagpur	2414	0.613 (0.588, 0.637)	2.90×10^{-249}

Statistical and practical significance: The extremely small p-values (< 0.001) confirm that these correlations are not due to random variation. However, the differing magnitudes across cities also have policy implications. A higher r value, as observed in Nagpur, signals limited affordability resilience—where any increase in travel time directly burdens users financially. Conversely, Bengaluru's moderate correlation reflects greater accessibility to low-cost or non-motorised options, highlighting the benefits of diversified last-mile mode availability.
Insight for implementation: Understanding the strength of this time–cost relationship can guide planners in prioritising equitable last-mile strategies. Cities with stronger coupling (high r) should focus on:

- Enhancing low-cost feeder and shared mobility services,
- Improving pedestrian and cycling infrastructure to reduce short-trip motorisation, and
- Introducing fare integration mechanisms that decouple user cost from trip length.

Overall, the correlation analysis underscores the economic sensitivity of last-mile accessibility, reinforcing that improving travel efficiency (reducing time) can yield not only comfort and reliability gains but also tangible affordability benefits.

3.4 Research questions

To guide the analysis, we posed the following research questions spanning behavioural (Power BI) and spatial (QGIS) dimensions of last-mile connectivity across Bengaluru, Delhi, and Nagpur:

Behavioural (Power BI)

1. What are the dominant last-mile modes in each city, and how do city-wise shares compare across NMT, IPT, PV, BUS, and MIXED?
2. Are mode choices associated with demographic attributes (gender, age, income), and if so, with what practical effect size?
3. Do mean last-mile travel times differ significantly across modes, and how large are the effects by city?
4. Which modes are fastest/slowest within each city?
5. Is there a relationship between last-mile travel time and last-mile cost?

Spatial (QGIS)

1. At the station level, what is the dominant last-mile mode and percentage of walking trips?
2. Which stations qualify as high non-walk dependency zones under the study thresholds (%Walk $< 65\%$ or TT_LM > 15 min)?
3. How do spatial patterns differ between core and peripheral areas, and how do cities compare on key accessibility indicators?

Chapter 4

Power BI and QGIS Analysis: Insights and Recommendations

4.1 Power BI Analysis

The Power BI workflow was developed to visualize and interpret last-mile travel behaviour across Bengaluru, Delhi, and Nagpur using the cleaned and processed datasets prepared in Sections 3.1 and 3.2. The entire process—from data import to dashboard design—was structured to ensure transparency, reproducibility, and policy relevance.

4.1.1 Data Import and Transformation

The finalized Excel files for the three cities were imported into Power BI using the **Power Query Editor**. Within Power Query, redundant and missing records were removed, distance units were standardized to kilometres, and categorical variables such as mode, income, and age were recoded based on the classifications defined earlier. All tables were checked for consistent field names to ensure smooth integration across visualizations.

4.1.2 Data Model and Relationships

Separate tables representing respondent information, station attributes, and city identifiers were linked using unique keys such as *Station Name* and *City*. This relational model enabled synchronized filtering of visuals—allowing every graph or map to update dynamically when a particular mode, city, or demographic segment was selected.

4.1.3 Dashboard Design and Visualizations

Three interactive dashboard pages were created, each focusing on a different analytical dimension of last-mile mobility. The principal visuals included:

- **Pie Charts:** Displayed mode share for each city to identify the dominant last-mile connection types.

- **Scatter Plots:** Represented the relationship between *travel cost* and *travel time*, coloured by mode, to highlight cost-efficiency differences across modes.
- **Histograms:** Illustrated the distributions of *waiting time* and *travel distance*, revealing variability in access and service reliability.
- **Maps:** Used Power BI's built-in map visual to present station-wise mode shares, allowing spatial comparison of accessibility patterns.
- **Tree Maps:** Summarized respondents' *reasons for mode choice*, enabling quick interpretation of behavioural motivations.
- **Box Plots:** Compared *travel time across modes* to identify dispersion and outliers.
- **Slicers:** Added for *age*, *gender*, *income category*, and *vehicle ownership*, providing instant demographic filtering.

All visuals were focused exclusively on **last-mile travel**, consistent with the project scope.

4.1.4 Interactivity and Navigation

Cross-filtering was enabled across all charts so that the selection of a mode (for instance, "Auto") automatically updated the scatter, histogram, and map visuals. **Bookmarks** and **navigation buttons** were introduced to allow users to switch seamlessly between thematic pages—"Mode Share Overview," "Cost–Time Analysis," and "Demographic Insights." Tooltips were customized to display key numerical values when hovering over data points, enhancing interpretability.

4.1.5 Insights Derived

The Power BI dashboard was designed not only as a visualization tool but also as an analytical interface. The major insights drawn from the visuals include:

- **Dominant Modes:** Identification of the most preferred last-mile modes for each city. And it is found that NMT i.e. cycling and walking are preferred as most commonly in all the cities and their percentage share is shown in the table below -

City →	Bangalore	Delhi	Nagpur
NMT share (%)	73.4	63.33	76.64

- **Demographic Influence:** Observation of how income and gender shape mode preference.
  - **Gender :** For all the cities respondents have equally likely mode preferences irrespective of their gender. Most common preferred mode is NMT and after that it varies city wise. In the table below we have shown top 3 preferred mode for all the cities irrespective of gender -

City →	Bangalore	Delhi	Nagpur
Top three preferred mode	* NMT * IPT * Bus + NMT	* NMT * IPT * IPT + NMT	* NMT * IPT + NMT * IPT

- **Income :** Like gender, irrespective of income for all the cities, respondents prefer NMT as their last mile mode. But their preference towards IPT varies and which is shown in the table below -

City →	Bangalore	Delhi	Nagpur
Income group wise % share of IPT from their overall mode share	* Low → 13.74% * Medium → 21.8% * High → 21.84%	* Low → 36% * Medium → 17.89% * High → 12.5%	* Low → 15.97% * Medium → 13.76% * High → 13.89%

→ The discrepancy between income group and their preference towards IPT for Delhi and Nagpur is mainly due to the high percentage share of low income category respondents.

- **Age :** All the respondents have their first priority as a last mile mode is NMT irrespective of their age as well and across all the cities. However, their preference varies towards IPT and which is shown in the table below -

City →	Bangalore	Delhi	Nagpur
Age group wise % share of IPT from their overall mode share	* Child → 12.05% * Adult → 15.18% * Old → 0%	* Child → 41.54% * Adult → 35.5% * Old → 11.11%	* Child → 14.29% * Adult → 13.87% * Old → 0%

→ Again the discrepancy of % share for old respondents in cities like Bangalore and Nagpur is mainly due to the lower percentage of old age respondents in the overall data.

- **Spatial Hotspots:** Detection of stations with the highest average last-mile time, and our findings are tabulated in the table below -

City →	Bangalore	Delhi	Nagpur
Station with highest average last mile time	Baiyappanhalli	Shivaji park	Kasturchand park
Highest average value of last mile time (minutes)	14.15	38	18.61

The resulting interactive dashboard thus enables planners and researchers to explore behavioural patterns in last-mile connectivity dynamically and to communicate findings effectively to decision-makers.

4.1.6 Recommendations (Based on Power BI Analysis)

The Power BI dashboard highlights distinct behavioral and infrastructural trends across Bengaluru, Delhi, and Nagpur. Based on the analysis, the following key recommendations are proposed:

1. **Strengthen Walking and Cycling Access:** Since a majority of last-mile trips are completed by walking or cycling, continuous, shaded, and obstruction-free footpaths with safe pedestrian crossings should be developed. Provision of short-distance bicycle-sharing facilities within 500 m of metro stations will further promote non-motorized access.

2. **Improve Station-Area Connectivity:** Stations exhibiting longer access times or low walking shares should be prioritized for redesign. Dedicated auto/e-rickshaw bays, mid-block crossings, and universally accessible routes will enhance multimodal integration and user convenience.

3. **Integrate Intermediate Public Transport (IPT):** Autos and shared IPT modes play a critical role for medium-distance commuters. Their operations should be formalized through designated pick-up zones, coordinated routes, and integrated fare systems to ensure smooth transfers without congestion.

4. **Ensure Equity and Inclusivity:** Low-income and female commuters rely heavily on IPT and walking. Safety improvements such as lighting, CCTV surveillance, and visible staff presence, along with subsidized feeder fares and accessible pathways in low-income neighborhoods, are essential for equitable mobility.

5. **Use Dashboards for Continuous Monitoring:** The Power BI dashboard should be institutionalized as a dynamic monitoring tool. Regular updates and performance tracking will help city agencies identify emerging accessibility gaps and assess the impact of policy interventions over time.

4.2 QGIS Process and Spatial Analysis

The QGIS workflow was developed to spatially represent and interpret the last-mile connectivity characteristics for Bengaluru, Delhi, and Nagpur. While Power BI was used to analyse behavioural and numerical aspects, QGIS enabled the mapping of spatial accessibility patterns — showing how the dominant mode, share of walking trips, and average last-mile travel time vary geographically across metro stations. Together, these analyses allowed a comprehensive understanding of both behavioural and physical dimensions of last-mile travel.

4.2.1 Data Preparation and Integration

The preparation of datasets for use in QGIS followed a consistent structure for all three cities:

- **Geocoding of Stations:** Metro station names were converted into geographic coordinates (latitude and longitude) with official metro shapefiles.
- **Layer Imports:** Shapefiles and GeoJSON files for road networks were imported from OpenStreetMap.
- **Attribute Join:** The cleaned Excel datasets containing variables such as `Dominant_LM_Mode`, `%WALK_LM_TRIPS`, and `TT_LM` were joined to the corresponding station point layers.
- **Projection and Formatting:** All data were standardised to WGS 84 (EPSG 4326) and symbolised using a common colour scheme to ensure comparability between the three cities.

4.2.2 Map Development and Layouts

Three mandatory maps and one optional map were created for each city as per the project brief:

1. **Mode Share by Station:** Colour-coded points indicating the dominant last-mile mode (NMT, IPT, or Other) at each station.
2. **Accessibility Buffers:** 500 m and 1 km buffers drawn around stations, shaded by the percentage of walk trips (`%WALK_LM_TRIPS`) to represent walk accessibility.
3. **Travel-Time Choropleth:** Average last-mile travel time (`TT_LM`) symbolised using a sequential blue gradient ranging from 7 to 18 minutes — darker shades indicating higher travel times.
4. **(Optional) Infrastructure Gap Map:** Overlays of existing footpaths and cycle tracks (`highway=footway`, `cycleway`) to identify missing links in non-motorised infrastructure.

Each map layout was designed with a clear title, north arrow, legend, and scale bar. The maps were exported as high-resolution `.pdf` files and arranged sequentially for each city in the report.

4.2.3 Analytical Approach

For all cities, three station-level variables were used to evaluate accessibility:

- Dominant_LM_Mode — the primary last-mile connection type,
- %Walk — share of last-mile trips completed by walking,
- TT_LM (min) — average last-mile travel time.

A station was categorised as a high non-walk dependency zone if either of the following conditions was satisfied:

$$\%\text{Walk} < 65\% \quad \text{or} \quad \text{TT_LM} > 15 \text{ minutes}.$$

This classification criterion was developed based on two complementary foundations:

- **Threshold for Walking Share:** As per international walkability and last-mile connectivity studies (e.g., ITDP, 2018; SUTP–GIZ, 2019), metro catchment areas where walking constitutes less than two-thirds of total access trips (<65%) are considered to exhibit high dependency on intermediate or motorised feeder modes. This 65% benchmark is consistent with findings from Indian metro systems (CSTEP, 2020; RITES, 2016), where walk shares typically range between 60–80% for well-connected stations.
- **Threshold for Travel Time:** The 15-minute travel-time cut-off is drawn from the *"15-Minute City"* accessibility principle (Moreno et al., 2021) and related public transport accessibility literature (Ewing & Cervero, 2010). Studies on first/last-mile access (Ghosh et al., 2019; ITDP, 2018) indicate that when the average walking time exceeds 12–15 minutes, users tend to switch to motorised feeders or private vehicles, marking a threshold for reduced pedestrian viability.

Hence, a walk share below 65% or an average last-mile travel time above 15 minutes was adopted as a practical and literature-supported criterion for identifying **high non-walk dependency zones** in this study.

4.2.4 Spatial Analysis for Nagpur

In Nagpur, 23 stations were analysed. The results show that Non-Motorised Transport (NMT) is the dominant mode across all stations, indicating a strong walking culture.

High Non-Walk Dependency Zones

Stations such as **Kasturchand Park**, **Airport**, and **Jai Prakash Nagar** exceeded the defined threshold. Kasturchand Park and Airport show moderate walking shares (68.5% and 75.0% respectively) and longer last-mile travel times (18.4 and 17.5 minutes). **Jai Prakash Nagar**, with a walk share of only 57.9% and an average travel time of 16.4 minutes, breaches both limits and represents the most critical non-walk-dependent station in the network. These results highlight the need for improved pedestrian pathways and reduced access time in peripheral areas of the city.

Spatial Observations

The central cluster (Sitabuldi–Zero Miles–Dharampeth–Congress) shows high walk shares (77–89 %) and moderate travel times (10–13 min), confirming good accessibility and compact land-use patterns. Peripheral stations such as Airport and Jai Prakash Nagar record longer access times and stronger IPT dependence.

Performance Summary

Table 4.1: Nagpur – Station-Level Last-Mile Performance Summary

Metric	Bottom 3 (Needs Attention)	Top 3 (Good Practice)
% Walk Share	• Jai Prakash Nagar (57.9) • Kasturchand Park (68.5) • Vasudev Nagar (69.8)	• Shankar Nagar (93.3) • Jhansi Rani (91.4) • New Airport (90.9)
Travel Time (min)	• Kasturchand Park (18.4) • Airport (17.5) • Jai Prakash Nagar (16.4)	• New Airport (7.9) • LAD Chauk (9.6) • Congress Nagar (10.1)

Recommendations

Improving pedestrian continuity and providing mid-block crossings can enhance accessibility. Stations such as Shankar Nagar and Jhansi Rani may be used as benchmarks for best-practice NMT design.

4.2.5 Spatial Analysis for Delhi

Delhi's metro network shows a strong core–periphery contrast. Central stations are compact and pedestrian-friendly, while outer-ring and highway-edge stations display higher dependency on IPT.

High Non-Walk Dependency Zones

Peripheral stations such as **Shivaji Park**, **Dwarka Mor**, and **Rohini West** show low walk shares (< 60 %) and high travel times (> 20 min), highlighting the need for better pedestrian and feeder integration.

Spatial Observations

Inner-core stations (Mandi House, Rajiv Chowk, Kashmere Gate) achieve 80–85 % walk shares with travel times of 10–13 minutes, while outer extensions beyond the Ring Road exhibit much longer access distances.

Performance Summary

Table 4.2: Delhi – Station-Level Last-Mile Performance Summary

Metric	Bottom 3 (Needs Attention)	Top 3 (Good Practice)
% Walk Share	• Shivaji Park (54) • Dwarka Mor (58) • Rohini West (59)	• Mandi House (85) • Kashmere Gate (83) • Rajiv Chowk (81)
Travel Time (min)	• Shivaji Park (38) • Rithala (32) • Dwarka Mor (30)	• Mandi House (10) • Kashmere Gate (12) • Rajiv Chowk (13)

Recommendations

Continuous sidewalks, frequent safe crossings, and co-located IPT stands are essential to reduce walking detours and delays. Outer-ring corridors should prioritise universal pedestrian design and wayfinding improvements.

4.2.6 Spatial Analysis for Bengaluru

Bengaluru's metro (Namma Metro) presents moderate to high walk shares in residential and mixed-use areas but lower accessibility near industrial corridors.

High Non-Walk Dependency Zones

Stations such as **Baiyappanahalli**, **Peenya Industrial Area**, and **Nagasandra** report TT $>$ 14 min and %Walk $<$ 65 %, showing poor pedestrian continuity.

Spatial Observations

Core stations like MG Road, Vidhana Soudha, and Jayanagar exhibit strong walking shares (84–90 %) and short travel times ($<$ 10 min). Peripheral areas beyond the Outer Ring Road display longer access times and higher IPT reliance.

Performance Summary

Table 4.3: Bengaluru – Station-Level Last-Mile Performance Summary

Metric	Bottom 3 (Needs Attention)	Top 3 (Good Practice)
% Walk Share	• Baiyappanahalli (61) • Peenya Industrial Area (63) • Nagasandra (64)	• Vidhana Soudha (90) • MG Road (87) • Jayanagar (84)
Travel Time (min)	• Baiyappanahalli (14.2) • Peenya (14.0) • Nagasandra (13.8)	• Vidhana Soudha (8.9) • MG Road (9.2) • Jayanagar (9.5)

Recommendations

Improving shaded, barrier-free sidewalks and relocating IPT stands closer to station entries can significantly reduce last-mile time. Industrial corridors require targeted NMT interventions with protected crossings and signage.

4.2.7 Cross-City Comparison and Key Insights

A comparative summary of accessibility indicators is presented in Table 4.4.

Table 4.4: Comparative Summary of Accessibility Indicators

Indicator	Nagpur	Delhi	Bengaluru
Stations with %Walk ≥ 80%	74%	63%	73%
Median TT (min)	12.5	21.0	14.0
High Non-Walk Stations (Count)	3 / 23	7 / 209	4 / 52
Dominant Mode (NMT Share)	76.6%	63.3%	73.4%

Interpretation

Nagpur exhibits the highest overall walkability with short average travel times and compact station catchments. Delhi shows the strongest disparity between central

and peripheral areas, while Bengaluru maintains consistent performance with a few industrial outliers.

Chapter 5

Conclusion

This project analyzed last-mile connectivity in Bengaluru, Delhi, and Nagpur by integrating behavioural dashboards (Power BI) with station-level spatial analysis (QGIS). The combined approach clarified who uses which modes, how long last-mile trips take, and where access gaps persist around metro stations, providing a consistent evidence base for planning and policy.

Key findings. (i) Non-motorized transport (NMT) dominates last-mile access in all three cities. (ii) Demographics have limited practical effects: mode choice is independent of gender in every city; age is largely independent except for a weak association in Delhi; income shows weak but statistically significant associations (strongest—yet still small—in Delhi). (iii) Inter-modal time differences are substantial: mixed/bus-linked trips are longest, NMT is consistently shortest; effects are large in Delhi and Nagpur and moderate in Bengaluru. (iv) Last-mile time and cost are positively correlated in every city, implying that reducing access time also improves affordability.

Spatial insights. QGIS mapping identified high non-walk dependency stations (e.g., %Walk $< 65\%$ or $\mathrm{TT}_{\mathrm{LM}} > 15$ min). Peripheral and industrial-corridor stations require targeted upgrades, while well-performing core stations provide transferable design practices. Cross-city comparison shows Nagpur's compact catchments and highest walkability; Delhi's strongest core–periphery contrast; and Bengaluru's steady performance with a few outliers.

Implications. Three priorities emerge: (1) protect and upgrade NMT first (continuous, shaded, barrier-free footpaths and safer crossings near every station), (2) integrate IPT as a complement (designated bays, closer curb access, coordination to reduce waiting/transfer times), and (3) target hotspots breaching walk/time thresholds rather than generic corridor-wide fixes.

Limitations and extensions. Findings reflect available survey data and categorisation choices; some modes were sparse and grouped as MIXED. Future work should add time-of-day effects, on-ground pedestrian audits, and fare/frequency integration to quantify benefits. Linking these datasets within the same Power BI–QGIS pipeline enables continuous monitoring and station-level benchmarking.

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